

23/24 Sem 2 Q2(1)

Given $f(0) = f(1) = g(0) = g(1) = 0$

Aim: $\int_0^1 f''(x) g(x) dx = \int_0^1 f(x) g''(x) dx.$

$$f''(x) \xrightarrow{\text{int}} f'(x) \xrightarrow{\text{int}} f(x)$$

$$g(x) \xrightarrow{\text{diff}} g'(x) \xrightarrow{\text{diff}} g''(x)$$

Method: Use int. by part

$$\int_0^1 \underbrace{f''(x)}_{\text{int}} \underbrace{g(x)}_{\text{diff}} dx$$

	diff	int
+	$g(x)$	$f''(x)$
-	$g'(x)$	$f'(x)$

$$= [g(x) f'(x)]_0^1 - \int_0^1 g'(x) f'(x) dx$$

(by part)

$$= \underbrace{g(1)}_{=0} f'(1) - \underbrace{g(0)}_{=0} f'(0) - \int_0^1 g'(x) f'(x) dx$$

$$= - \int_0^1 \underbrace{g'(x)}_{\text{diff}} \underbrace{f'(x)}_{\text{int}} dx$$

$$= - \left([g'(x)f(x)]_0^1 - \int_0^1 g''(x)f(x) dx \right)$$

(by part)

$$= - \left(\underbrace{g'(1)f(1)}_{=0} - \underbrace{g'(0)f(0)}_{=0} - \int_0^1 g''(x)f(x) dx \right)$$

$$= - \left(- \int_0^1 g''(x)f(x) dx \right)$$

$$= \int_0^1 g''(x)f(x) dx$$
